

Richard van Zyl — Skills Overview

Solutions Architect / Technical Lead — Backend & Data-Intensive Systems

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Years column. Professional years only, counted from paid work starting **April 2017** — nothing exceeds **9**. Study years (2014–2016) are not counted.

At a glance

Area	Strength
Primary stack	C# / .NET, SQL Server / Azure SQL, T-SQL
Architecture	Multi-tenant financial systems,DDD, CQRS, idempotent ledger design
Cloud	Microsoft Azure in production (not AWS)
Delivery	Azure DevOps YAML pipelines, Docker on Azure hosts, trunk-based + feature flags
Leadership	~5 years leading teams; ~3.5 as Solutions Architect / Technical Lead
Analysis	BRS / URS / FRS / TRS since 2017; stakeholder translation

Grouped skills (professional years)

Languages & data formats

Skill	Years
C# / .NET	9
T-SQL	9
PostgreSQL	7
TypeScript / JavaScript	5
JSON / XML	9 / 8
YAML	6
Visual Basic .NET	3
Java	2

Databases & data engineering

Skill	Years
SQL Server / SSMS	9
SQL performance tuning	8
Azure SQL	7
Entity Framework	9
Redis (distributed caching)	5
Table partitioning & archiving	4

Skill	Years
In-Memory OLTP	4
MongoDB	4
SSRS	5

Architecture & design

Skill	Years
Clean Architecture	9
Domain-Driven Design	8
Microservices	7
CQRS	6
Event-driven architecture	5
Multi-tenancy / tenant isolation	7
Idempotency / exactly-once processing	4

Requirements & solution design

Skill	Years
Business & problem-domain analysis	9
Requirements (BRS / URS / FRS / TRS)	9
Solution design & architecture	9
Technical documentation (audience-tailored)	9
Stakeholder engagement & translation	8
Process & system modelling (UML)	8

Backend frameworks & practices

Skill	Years
MediatR / AutoMapper / FluentValidation	7 / 6 / 6
OAuth2 / OpenID Connect / JWT	6
Unit testing (xUnit / NUnit / Moq)	6
Integration testing (TestContainers)	5
Feature flags (Azure App Configuration)	4
Circuit breaker (Polly)	3
SignalR / WebSockets	3
Angular / Angular Material	4
React	4

Cloud, DevOps & observability

Skill	Years
Azure DevOps (previously TFS)	9
Azure CI/CD	7

Skill	Years
Pipeline-as-code (Azure DevOps YAML)	6
Docker	7
Message queues (Azure Service Bus / RabbitMQ)	6
Azure Monitor / App Insights / Log Analytics	5
New Relic	4
API Gateway (Azure API Management)	4
OpenTelemetry / Grafana / Graylog	3

Security & compliance

Skill	Years
Role-Based Access Control (RBAC)	8
Data-protection-aware design (POPIA / GDPR)	5
SOX-aligned data segregation	4

Honest boundaries

Keep these framing notes when matching roles — they are deliberate, not soft-peddling.

- **No AWS experience.** Cloud depth is **Microsoft Azure**. Do not map skills to “AWS equivalents.”
- **Infrastructure as Code:** declarative work is **Azure DevOps YAML pipelines** (pipeline-as-code) and **Docker on Azure hosts** — not Terraform, Bicep, or ARM templates in anger. Roles that lead on cloud resource provisioning as code need ramp-up; that is not a current strength.
- **Python:** not a language to hire for. Production exposure is **IronPython hosted in a C# backend** for integration marshalling (pre-v3), plus limited contemporaneous Python at MeterMo. Treat as read-and-modify.
- **Front-end:** not the specialisation. Angular with React ran **Apr 2020 - Dec 2022** (Payteq, Dotcom, MeterMo), React hands-on alongside Angular on each of those engagements rather than observed from the next desk. Since then the front-end is *directed rather than typed* - specification, review and architecture, with agents doing the implementation. That is a current working posture, not a lapsed skill. What carries across is type discipline: a long JavaScript foundation made the move to TypeScript immediate, and the draw was that it can be made *fully* strictly typed - `strict: true`, no implicit any, enforced at the linter rather than left to convention - the same instinct as the .NET strict-mode work. For a team that needs its front-end owned end to end by the person leading it, this is not the right fit; for one that needs it specified, reviewed and held to a standard, it is.
- **.NET version currency:** employment history ran on the .NET Core and .NET 6 line; current active development is on **.NET 10** (CombinatorialOptimiser). The step between them is a version upgrade, not a re-skill - and the evidence for handling exactly that is the BotFramework migration from .NET Standard to .NET 6, completed and shipped after earlier attempts had been rolled back.
- **Docker / containerisation:** production use of Docker on Azure hosts and containerised platforms. Cluster orchestration is not a current skill — would rely on operational support for that layer.
- **ActiveXperts:** used in production observability prose; not given a years row here (no invented span).

SQL performance tuning (condensed)

~8 years on high-volume financial ledger systems (150+ money processors across African markets).

Focus	What it means in practice
Execution plans	SHOWPLAN, STATISTICS IO / TIME; spotting scans, lookups, high-cost operators
Indexing	Clustered / non-clustered / filtered / covering; fragmentation and rebuild choices
Locking & concurrency	Isolation levels; deadlock detection via Profiler / Extended Events
Tuning patterns	SARGable predicates; join vs EXISTS; avoiding scalar UDFs; subquery rewrites
Scale design	Date/tenant partitioning; indexed/materialised views; high-volume batch inserts
In-Memory OLTP	Memory-optimised tables and natively compiled procedures on hot paths
Health monitoring	Query Store, wait stats, index usage, sp_WhoIsActive

Career timeline (condensed)

Period	Role	Organisation
Apr 2026 – present	Stakeholder / Solutions Architect (<i>side project</i>)	Yuno Technologies (partner-operated)
current	Co-founder (<i>side project</i>)	Orchestration Platform
Dec 2022 – Mar 2026	Solutions Architect / Technical Lead — Core Financial Systems	Raging River Trading
Apr 2022 – Dec 2022	Software Developer	MeterMo (utilities metering; six projects TFS→Azure DevOps; APIs, field apps, websites; platform-debt close)
Jan 2022 – Apr 2022	Software Developer (Specialist Problem-Solver)	Dotcom Software Solutions
Apr 2020 – Jan 2022	Senior Developer / Business Analyst / Team Lead	Payteq
Oct 2017 – Apr 2020	Software Developer / Business Analyst	iPlan Global
Apr 2017 – Oct 2017	Software Developer (C# .NET)	Novigo

Education & assessment

- **National Diploma — IT: Software Development**, Varsity College, 2014–2016
- **IKM C# Assessment**: 73rd percentile (assessed twice, six months apart)

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